

AWS DevOps Engineer Intern Assignment Report

Submitted By: Priyanshu Singh

College: VIT Bhopal University

Branch: B.Tech CSE (Cyber Security & Digital Forensics)

Date: 09 July 2026

1. Objective

The objective of this assignment was to deploy a simple static website on an AWS EC2 Ubuntu instance using the Nginx web server. The assignment also involved configuring networking, performing essential Linux administration tasks, managing the project with Git and GitHub, and documenting the complete deployment process.

2. AWS Services Used

Amazon EC2

Amazon Elastic Compute Cloud (EC2) was used to launch a virtual Ubuntu server that hosted the static website.

Purpose

- Launch a Linux virtual machine
- Host the website
- Perform Linux administration tasks

Security Groups

AWS Security Groups were configured as a virtual firewall for the EC2 instance.

Inbound Rules

Port	Protocol	Purpose
22	SSH	Secure remote login
80	HTTP	Website access

Key Pair

An AWS Key Pair was generated and used for secure SSH authentication to the EC2 instance.

3. Environment Details

Component	Value
Operating System	Ubuntu 22.04 LTS
Cloud Platform	AWS
Web Server	Nginx
Version Control	Git
Repository Hosting	GitHub
Website Type	Static HTML

4. Linux Commands Used

Update Package Index

```
sudo apt-get update
```

Updates the local package repository information.

Upgrade Installed Packages

```
sudo apt upgrade -y
```

Installs the latest available package updates.

Install Nginx

```
sudo apt install nginx -y
```

Installs the Nginx web server.

Check Nginx Status

```
sudo systemctl status nginx
```

Verifies that the Nginx service is running correctly.

Restart Nginx

```
sudo systemctl restart nginx
```

Restarts the web server after configuration changes.

Enable Nginx at Boot

```
sudo systemctl enable nginx
```

Ensures Nginx starts automatically after a reboot.

Check Disk Usage

```
df -h
```

Displays available and used disk space.

Check Memory Usage

```
free -h
```

Displays RAM and swap memory usage.

Check Running Processes

```
top
```

Displays currently running system processes.

5. Website Deployment Process

1. Created an Ubuntu EC2 instance.
2. Configured Security Group rules to allow SSH (22) and HTTP (80).
3. Connected to the instance using SSH.
4. Updated system packages.
5. Installed and configured the Nginx web server.
6. Created a custom HTML landing page.
7. Replaced the default Nginx web page with the custom page.
8. Restarted the Nginx service.
9. Verified website accessibility through the EC2 Public IP.
10. Uploaded project files to GitHub.

6. Git and GitHub Workflow

The project source code was managed using Git and uploaded to GitHub.

Commands Used

```
git init
git add index.html
git add README.md
git commit -m "Initial AWS DevOps assignment"
git branch -M main
git remote add origin https://github.com/PriyanshuSingh10114/aws-devops-intern-assignment.git
git push -u origin main
```

7. Bonus Tasks Completed

Docker Installation

Docker was installed on the EC2 instance and verified successfully by running the official hello-world container.

Commands

```
sudo apt install docker.io -y
sudo systemctl enable docker
sudo systemctl start docker
sudo docker run hello-world
```

8. Challenges Faced

During the assignment, the following challenges were encountered:

- Configuring Security Group rules correctly to allow website access.
- Understanding the default Nginx web root directory.
- Setting up SSH authentication for GitHub from the EC2 instance.
- Verifying correct file permissions while replacing the default web page.
- Organizing project documentation and screenshots for submission.

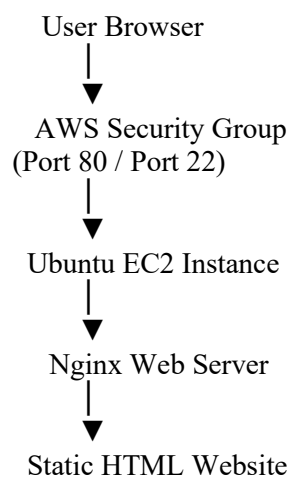
Each issue was resolved by referring to official documentation and verifying the configuration step by step.

9. Learning Outcomes

This assignment helped strengthen practical knowledge in:

- AWS EC2 instance provisioning
- Linux system administration
- SSH authentication
- Nginx installation and configuration
- Static website deployment
- Git and GitHub version control
- Docker fundamentals
- Technical documentation and reporting

10. Project Architecture



11. Time Taken

Task	Time
EC2 Setup	3 Minutes
Linux Configuration	3 Minutes
Website Deployment	3 Minutes
Git & GitHub	5 Minutes
Docker Bonus	3 Minutes
Documentation	30 Minutes

Total Time: Approximately 47 Minutes

12. Conclusion

This assignment provided practical experience in deploying and managing a web application on AWS using industry-standard tools. It strengthened my understanding of cloud infrastructure, Linux administration, web server configuration, version control, containerization, and automation. The hands-on experience reinforced DevOps fundamentals and demonstrated the complete workflow of deploying and documenting a cloud-hosted application.